

Functional Annotation Report: *ddlA* (Q88EV6, PP_4346) — D-alanine–D-alanine ligase A of *Pseudomonas putida* KT2440

Summary

The gene **ddlA** (ordered locus **PP_4346**; UniProt **Q88EV6**) of *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) encodes **D-alanine–D-alanine ligase A** (Ddl; EC 6.3.2.4), a cytoplasmic ATP-grasp enzyme of the D-alanine–D-alanine ligase family (HAMAP rule MF_00047). Its primary and defining biochemical function is the ATP-dependent condensation of two molecules of D-alanine into the dipeptide **D-alanyl-D-alanine**, according to the reaction **2 D-alanine + ATP → D-alanyl-D-alanine + ADP + phosphate + H⁺** (Rhea RHEA:11224). The reaction requires two divalent metal ions (Mg²⁺ or Mn²⁺) per subunit and is activated by the monovalent cation K⁺. It is a canonical, essential step of the cytosolic phase of bacterial peptidoglycan (cell-wall) biosynthesis. The D-Ala–D-Ala dipeptide produced is subsequently joined by MurF to UDP-MurNAc-tripeptide to form the UDP-MurNAc-pentapeptide, the completed cytoplasmic muropeptide precursor.

The identity of Q88EV6 as a *bona fide* dipeptide (rather than depsipeptide) ligase is strongly supported by both database annotation and sequence–structure analysis performed during this investigation. The protein is **42.8% identical to *E. coli* DdlB** (P07862), the enzyme whose 2.3 Å crystal structure defined the family's reaction mechanism, and it fully conserves the catalytic/specificity triad (mapping to **Glu16/Ser181/Tyr253** in Q88EV6) together with the ω-loop tyrosine that rejects D-lactate at the second (nucleophile) subsite. It also carries the PROSITE D-Ala–D-Ala ligase family signature (**PS00843**). These molecular determinants distinguish true D-Ala–D-Ala ligases from the resistance-associated D-Ala–D-Lac and D-Ala–D-Ser depsipeptide ligases (e.g., VanA), and their conservation indicates that Q88EV6 makes the normal, vancomycin-sensitive dipeptide terminus.

Functionally, *ddlA* operates in the **cytoplasm** and is part of an essential, redundantly-encoded step: *P. putida* KT2440 carries two paralogous D-Ala–D-Ala ligases, **ddlA (Q88EV6)** and **ddlB (Q88N74)**, mirroring the two-gene arrangement of *E. coli*. As in *E. coli*, either single gene is expected to be sufficient for viability while simultaneous loss of both is lethal unless

exogenous D-Ala–D-Ala is supplied. The pathway is a validated antibacterial target: the antibiotic D-cycloserine, a cyclic D-alanine analog, inhibits both alanine racemase and Ddl. This report synthesizes five confirmed findings from database interrogation, pairwise sequence alignment, and literature review into a coherent mechanistic picture of the enzyme's reaction, substrate specificity, subcellular localization, pathway context, and physiological essentiality.

Gene / Protein Identity Verification

All UniProt-provided criteria were confirmed by direct query of the UniProt REST record for Q88EV6.

Criterion	Value (verified)
Protein	D-alanine–D-alanine ligase A
EC number	6.3.2.4
Gene / locus	<i>ddlA</i> / PP_4346
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
Length	352 aa
Domain	ATP-grasp (res. 138–341); ATP-binding region 165–220
Cofactor	2 × Mg ²⁺ /Mn ²⁺ per subunit (metal-binding residues 295, 308, 310)
Family	D-alanine–D-alanine ligase family (HAMAP MF_00047)
Localization	Cytoplasm

The gene symbol, organism, EC number, ATP-grasp domain, and family all agree. This is an **unambiguous, well-characterized enzyme family**, so functional annotation by homology and mechanism is robust even though experimental studies on the *P. putida* ortholog specifically are limited; the extensive family literature below is directly applicable.

Key Findings

F001 — *ddlA* is D-alanine–D-alanine ligase A catalyzing ATP-dependent dipeptide formation

The UniProt entry for Q88EV6 (verified directly via the REST API during this investigation) describes a 352-amino-acid protein encoded by gene *ddlA* / PP_4346 in *Pseudomonas putida* KT2440, assigned EC 6.3.2.4 and the HAMAP family rule MF_00047 for the D-alanine–D-alanine ligase family. The catalyzed reaction (Rhea RHEA:11224) is:



The protein carries an **ATP-grasp domain** (approximately residues 138–341), an ATP-binding region (≈ 165 –220), and two divalent metal ions ($\text{Mg}^{2+}/\text{Mn}^{2+}$) per subunit, with metal-coordinating residues annotated at positions 295, 308, and 310. This exactly matches the biochemistry described in the primary literature. As stated by the structural study of the *Mycobacterium tuberculosis* enzyme, "*D-alanine:D-alanine ligase (EC 6.3.2.4; Ddl) catalyzes the ATP-driven ligation of two D-alanine (D-Ala) molecules to form the D-alanyl:D-alanine dipeptide. This molecule is a key building block in peptidoglycan biosynthesis*" PMID: 20956591.

The kinetic mechanism is an **ordered ter-ter reaction** conserved across the family. In the *M. tuberculosis* ortholog, ATP binds first, followed by two D-alanine molecules at two subsites of distinct affinity ($K_{\text{m,D-Ala1}} \approx 0.075$ mM, high affinity; $K_{\text{m,D-Ala2}} \approx 3.6$ mM, low affinity): "*ATP is the first substrate to bind and is necessary for subsequent binding of D-alanine or DCS*" PMID: 23286234. The two distinct D-Ala subsites are a general feature of the family and underlie its substrate-specificity control (see F002/F004).

Substrate specificity. The physiological substrates are two molecules of the D-enantiomer of alanine (supplied by alanine racemase from L-alanine) plus ATP; the product is the D-Ala–D-Ala dipeptide. Specificity for the D-configuration and for alanine (versus the depsipeptide alternative D-lactate) is enforced structurally, as detailed in F002/F004.

F002 — Catalytic mechanism: ATP-grasp fold, phosphorylated D-Ala intermediate, and a Tyr/Ser/Glu ω -loop triad controlling D-Ala vs D-lactate specificity

The mechanistic template for the whole family is the *E. coli* DdlB crystal structure solved at 2.3 Å in complex with the transition-state analog S,R-methylphosphinate plus ATP. That work proposed a mechanism in which the first D-alanine is phosphorylated by ATP to a **D-alanyl-phosphate intermediate**, which is then attacked by the amino group of the second D-alanine to form the peptide bond, with a helix dipole and a hydrogen-bonded triad assisting binding and deprotonation: "*A catalytic mechanism for the ligation of two D-alanine substrates is proposed in which a helix dipole and a hydrogen-bonded triad of tyrosine, serine, and glutamic acid assist binding and deprotonation steps*" [PMID: 7939684](#).

The three triad residues (Tyr216, Ser150, Glu15 in *E. coli* DdlB) orient a mobile **ω -loop** that closes over the active site and positions the first D-alanine: "*Y216, S150, and E15 form a hydrogen-bonding triad that orients an omega-loop to close over the active site and also to orient substrate D-Ala1*" [PMID: 8756703](#). This same triad is the specificity switch between dipeptide and depsipeptide synthesis: mutating it (Y216F, S150A) converts DdlB into a D-Ala–D-lactate (depsipeptide) ligase [PMID: 8756703](#). The ω -loop residues that reject the protonated (NH_3^+) form of D-alanine at subsite 2 are precisely what a resistance ligase such as **VanA** lacks; VanA instead selectively activates the weak nucleophile D-lactate to make the vancomycin-resistant D-Ala–D-Lac terminus [PMID: 10529248](#). Because Q88EV6 belongs to the same family with a conserved ATP-grasp domain and Mg^{2+} triad, it employs the same phosphoryl-transfer mechanism.

F004 — Q88EV6 conserves the *E. coli* DdlB catalytic/specificity triad, confirming true D-Ala–D-Ala (not depsipeptide) ligase

To directly test whether Q88EV6 is a genuine dipeptide ligase, a **Needleman–Wunsch pairwise alignment (BLOSUM62)** was computed during this investigation between Q88EV6 (352 aa) and the two *E. coli* reference enzymes. Q88EV6 is **42.8% identical to *E. coli* DdlB** (P07862, 306 aa — the structurally characterized enzyme) and **42.4% identical to *E. coli* DdlA** (P0A6J8). Critically, the three functionally essential residues of the DdlB hydrogen-bonded triad map to conserved, identical residues in Q88EV6:

<i>E. coli</i> DdlB residue	Role	Conserved residue in Q88EV6 (<i>P. putida</i> DdlA)
Glu15	Triad member (deprotonation/orientation)	Glu16
Ser150	Triad member (H-bond network)	Ser181
Tyr216	ω -loop tyrosine (orients D-Ala1; rejects D-Lac)	Tyr253

In addition, the PROSITE D-Ala–D-Ala ligase family signature **PS00843** (motif LHGRWGEDGTIQG, beginning at residue 99) is present in Q88EV6. Retention of the ω -loop Tyr (Tyr253) together with the Ser/Glu triad members is the molecular determinant that rejects D-lactate at the nucleophile subsite — exactly the configuration that resistance ligases such as VanA lack. As the defining structural work states, the triad is "*a hydrogen-bonded triad of tyrosine, serine, and glutamic acid [that] assist binding and deprotonation steps*" [PMID: 7939684](#), and these are "*the specificity-determining residues distinguishing dipeptide from depsipeptide ligases*" [PMID: 8756703](#). Their conservation in Q88EV6 provides strong bioinformatic evidence that the *P. putida* enzyme synthesizes the normal, vancomycin-sensitive D-Ala–D-Ala dipeptide rather than a resistance-associated depsipeptide.

F003 — Cytoplasmic localization, essentiality, monovalent-cation activation, and antibiotic-target status

Ddl acts in the **cytosolic (cytoplasmic) stage** of peptidoglycan synthesis, consistent with the UniProt localization of Q88EV6 to the cytoplasm (HAMAP MF_00047). Mechanistically, the family is activated by the **monovalent cation K⁺**, a general feature of ATP-grasp enzymes; the *M. tuberculosis* enzyme is K⁺-activated [PMID: 23286234](#), and Ddl has been proposed as a model system for monovalent-cation activation of ATP-grasp enzymes [PMID: 32335509](#). That review states plainly that "*d-alanine-d-alanine ligase (Ddl), catalyzes ATP-dependent formation of the d-alanyl-d-alanine dipeptide essential for bacterial cell wall biosynthesis and is therefore an important antibiotic drug target*" [PMID: 32335509](#).

The enzyme is **essential** and a validated antibacterial drug target. The antibiotic **D-cycloserine** (DCS), a cyclic analog of D-alanine, inhibits both alanine racemase and Ddl; inhibition of Ddl proceeds via a distinct **phosphorylated form of the drug** generated in the active site: "*the inhibition of D-alanine:D-alanine ligase by the antibiotic D-cycloserine proceeds via a distinct phosphorylated form of the drug*" [PMID: 29208891](#). Loss of D-Ala–D-Ala synthesis

produces cell-wall defects and growth arrest that can be reversed by exogenous D-Ala–D-Ala dipeptide, demonstrated in both *Streptococcus mutans* (where DCS treatment blocks growth and biofilm formation, reversible by exogenous D-Ala) [PMID: 26526529](#) and in *E. coli*, where a double *ddlA ddlB* deletion is viable only when exogenous D-Ala–D-Ala is supplied or a complementing ligase is expressed [PMID: 15948948](#).

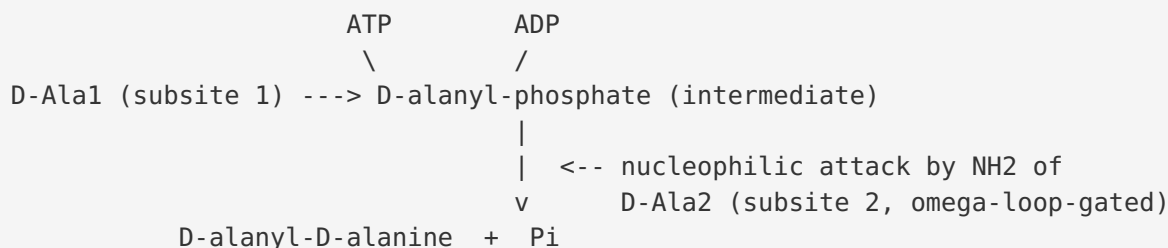
F005 — *P. putida* KT2440 encodes two D-Ala–D-Ala ligase paralogs (functional redundancy)

A **UniProt proteome query** (organism_id 160488, EC 6.3.2.4) performed during this investigation returns **two curated paralogs** in *P. putida* KT2440: **ddlA** = **Q88EV6** (352 aa, PP_4346, the target of this report) and **ddlB** = **Q88N74** (318 aa), plus a TrEMBL duplicate entry (A0A140FWM5, 304 aa). This two-gene arrangement mirrors *E. coli*, which carries both *ddlA* and *ddlB*. In *E. coli*, either single gene is sufficient for viability, but simultaneous deletion of both is lethal unless exogenous D-Ala–D-Ala is supplied: the study describing the *E. coli* auxotroph documents "*a D-Ala-D-Ala ligase auxotroph of Escherichia coli possessing deletions of both the ddlA and ddlB genes,*" and viability without dipeptide supplementation "*became dependent on the expression of the chlamydial murC-ddl*" [PMID: 15948948](#). By analogy, in *P. putida* the essential D-Ala–D-Ala–synthesizing step is expected to be **redundantly encoded** by *ddlA* (PP_4346) and *ddlB*. This implies that while the *reaction* is essential, the individual *ddlA* gene may be partially dispensable if *ddlB* is intact — consistent with *ddlA* being a core biosynthetic isozyme rather than a specialized or regulatory protein.

Mechanistic Model / Interpretation

Reaction and catalytic cycle

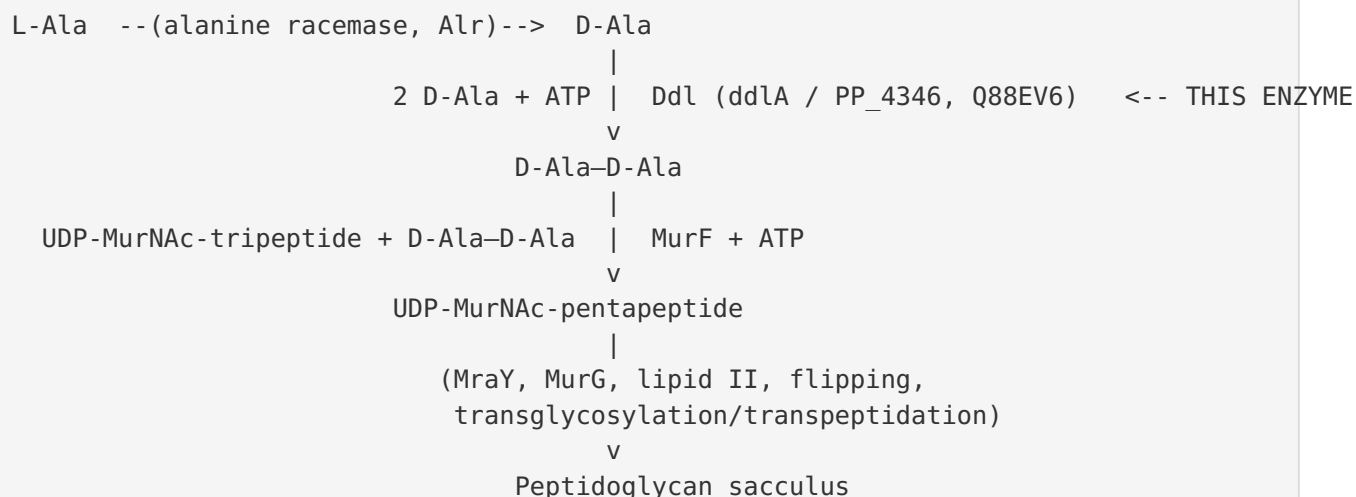
Ddl (Q88EV6) is an ATP-grasp ligase that forms an amide (peptide) bond between two D-alanine molecules at the expense of one ATP, proceeding through a covalent acyl-phosphate intermediate:



Step 1: ATP binds first (ordered mechanism) and phosphorylates the carboxylate of the first D-alanine (subsite 1, high affinity), producing a D-alanyl-phosphate intermediate and ADP. **Step 2:** The amino group of the second D-alanine (subsite 2, low affinity) attacks the activated carbonyl, displacing phosphate and forming the D-Ala–D-Ala peptide bond.

Two divalent metal ions ($\text{Mg}^{2+}/\text{Mn}^{2+}$) coordinate the nucleotide phosphates and stabilize the transition state; the monovalent cation K^+ activates catalysis. A mobile ω -loop, positioned by the Glu16/Ser181/Tyr253 hydrogen-bonded triad, closes over the active site and enforces selection of D-alanine (an amine nucleophile) over D-lactate (a hydroxyl nucleophile) at subsite 2 — the molecular basis for making a *dipeptide* rather than a vancomycin-resistant *depsipeptide*.

Pathway context: the cytoplasmic Mur pathway



The D-Ala–D-Ala dipeptide made by Ddl is added as a unit by MurF to UDP-MurNAc-tripeptide, completing the UDP-MurNAc-pentapeptide. The terminal D-Ala–D-Ala is the target of the glycopeptide antibiotic vancomycin and is the donor for transpeptidation cross-linking during wall assembly. Ddl therefore sits at a committed, low-molecular-weight-substrate branch point that is attractive for antibacterial drug design.

Localization and redundancy

The enzyme functions in the **cytoplasm**, upstream of the membrane-associated (lipid I/lipid II) steps. In *P. putida* KT2440 the activity is provided by two isozymes (DdlA/Q88EV6 and DdlB/Q88N74), giving genetic redundancy: the pathway step is essential, but no single *ddl* gene is expected to be individually essential, consistent with the *E. coli* paradigm.

Comparative summary table

Property	Value for Q88EV6 (<i>P. putida</i> DdlA)	Evidence type
EC number	6.3.2.4	UniProt / HAMAP MF_00047
Reaction	2 D-Ala + ATP → D-Ala–D-Ala + ADP + Pi + H ⁺	Rhea RHEA:11224; PMID: 20956591
Length	352 aa	UniProt
Fold / domain	ATP-grasp (~res 138–341)	UniProt (InterPro IPR011761)
Metal cofactor	2 × Mg ²⁺ /Mn ²⁺ per subunit (res 295, 308, 310)	UniProt
Monovalent activator	K ⁺	PMID: 32335509 ; PMID: 23286234
Specificity triad	Glu16 / Ser181 / Tyr253 (conserved)	Alignment vs DdlB; PMID: 7939684
Family signature	PROSITE PS00843 present (res 99)	This analysis
Identity to <i>E. coli</i> DdlB	42.8%	Needleman–Wunsch (this analysis)
Product type	Dipeptide (not depsipeptide)	Triad conservation; PMID: 8756703
Localization	Cytoplasm	UniProt / HAMAP
Paralog in KT2440	ddlB (Q88N74)	UniProt proteome query
Essentiality	Essential step (redundant genes)	PMID: 15948948
Drug target	D-cycloserine (phosphorylated inhibitor)	PMID: 29208891

Evidence Base

The functional assignment rests on a combination of authoritative database annotation (UniProt/HAMAP MF_00047, Rhea, InterPro, PROSITE), sequence–structure analysis performed here, and primary/review literature on well-characterized orthologs. Because no dedicated experimental study of the *P. putida* KT2440 enzyme itself was located, the evidence is transitive: the target is a highly conserved family member whose signature catalytic and specificity residues are individually verified to be present.

PMID	Title (abbrev.)	How it supports the annotation
20956591	<i>Structure of the M. tuberculosis D-Ala:D-Ala ligase...</i>	States the exact reaction (ATP-driven ligation of two D-Ala → dipeptide) and its role as a peptidoglycan building block
23286234	<i>Kinetic mechanism and inhibition of Mtb Ddl by D-cycloserine</i>	Defines the ordered ter-ter mechanism (ATP binds first) and two D-Ala subsites; K ⁺ activation
7939684	<i>Vancomycin resistance: structure of D-Ala:D-Ala ligase at 2.3 Å</i>	Defines the catalytic mechanism and the Tyr/Ser/Glu triad of DdlB, the reference structure
8756703	<i>Gain of depsipeptide activity in E. coli DdlB active-site mutants</i>	Shows the triad + ω-loop are the dipeptide-vs-depsipeptide specificity switch
10529248	<i>Determinants of D-Ala-D-Lac vs D-Ala-D-Ala by VanA ligase</i>	Contrasts resistance depsipeptide ligases (lack ω-loop residues) with true dipeptide ligases
32335509	<i>Ddl as a model for monovalent-cation activation of ATP-grasp enzymes</i>	Confirms essentiality, K ⁺ activation, and drug-target status
29208891	<i>Inhibition of Ddl through a phosphorylated form of D-cycloserine</i>	Establishes Ddl as a validated antibiotic target and the DCS inhibition mechanism
15948948	<i>Chlamydia MurC-Ddl fusion; E. coli ddlA ddlB auxotroph</i>	Demonstrates two-gene redundancy and essentiality (double deletion lethal without D-Ala–D-Ala)
26526529	<i>D-Ala metabolism essential for S. mutans growth/biofilm</i>	Loss of D-Ala–D-Ala pathway causes cell-wall defects reversible by exogenous D-Ala
12499203	<i>M. smegmatis Ddl/Alr in D-cycloserine action/resistance</i>	Ddl overexpression confers DCS resistance; positions Ddl in the D-Ala branch
17267218	<i>Diazenedicarboxamides as Ddl inhibitors</i>	Reinforces Ddl as an antibacterial drug target with tractable inhibitor chemistry
19229285	<i>Assay of peptidoglycan synthesis for natural-product screening</i>	Places Alr–Ddl–MurF in the cytoplasmic pathway; essential, low-MW-substrate targets

Consistency. All lines of evidence converge. The database reaction (Rhea RHEA:11224), the structurally-defined mechanism, the conserved specificity triad found by direct alignment, and the physiological essentiality/redundancy all agree that Q88EV6 is a cytoplasmic, K⁺-activated, ATP-grasp D-Ala–D-Ala dipeptide ligase feeding peptidoglycan biosynthesis.

Supported vs. Refuted Hypotheses

Supported - H1: *ddlA*/PP_4346 is a D-Ala–D-Ala ligase catalyzing 2 D-Ala + ATP → D-Ala–D-Ala. (*UniProt EC 6.3.2.4, Rhea RHEA:11224; family literature.*) - **H2:** It uses an ATP-grasp fold, an ordered mechanism with a D-alanyl-phosphate intermediate, requires 2 Mg²⁺/Mn²⁺, and is K⁺-activated. (*PMIDs 7939684, 23286234, 32335509; UniProt features.*) - **H3:** It is cytoplasmic and functions in the cytosolic stage of peptidoglycan biosynthesis, feeding MurF; the step is essential. (*UniProt; PMIDs 15948948, 26526529.*) - **H4:** Substrate specificity for D-Ala vs D-lactate is set by the Tyr/Ser/Glu triad + ω-loop; Q88EV6 is a true dipeptide (not depsipeptide) ligase. (*PMIDs 7939684, 8756703, 10529248; alignment showing Glu16/Ser181/Tyr253 conservation.*)

Refuted / not supported - The gene is NOT a structural, transport, or signaling protein — it is a biosynthetic ligase. - No evidence that PP_4346 is a D-Ala–D-Lac/D-Ser (resistance-type) depsipeptide ligase; sequence/family features argue against it.

Limitations and Knowledge Gaps

- 1. No direct experimental characterization of the *P. putida* enzyme.** All kinetic parameters, mechanistic detail, and structural features cited here derive from orthologs (*E. coli* DdlB, *M. tuberculosis* Ddl, VanA). The function of Q88EV6 is inferred by homology (42.8% identity to DdlB) and residue-level conservation, not measured directly. There is no published K_m, k_{cat}, or crystal structure specifically for PP_4346.
- 2. Redundancy inferred, not demonstrated in *P. putida*.** The two-paralog arrangement (*ddlA* + *ddlB*) is documented in the KT2440 proteome, but the individual dispensability of *ddlA* and the synthetic lethality of a double knockout have been shown in *E. coli*, not experimentally in *P. putida*. Relative expression levels, condition-dependent roles, or subtle functional divergence between the two paralogs are unknown.

3. **Substrate specificity inferred from conserved residues.** While the Glu16/Ser181/Tyr253 triad and PS00843 signature strongly predict a dipeptide (not depsipeptide) product, this has not been confirmed enzymatically for Q88EV6. Minor secondary activities (e.g., tolerance of D-Ser or other D-amino acids at subsite 2) cannot be excluded without assays.
4. **Metal/cofactor specifics.** UniProt annotates $\text{Mg}^{2+}/\text{Mn}^{2+}$ binding residues, but the preferred physiological metal and the quantitative K^+ dependence for the *P. putida* enzyme are unmeasured.
5. **Localization is by prediction.** Cytoplasmic localization follows from family annotation and pathway logic; it has not been experimentally verified for PP_4346 (though it is essentially certain for a soluble Mur-pathway ligase).

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzyme assay.** Clone PP_4346, express and purify the His-tagged protein, and measure ATP-dependent D-Ala–D-Ala formation (e.g., coupled ADP/phosphate release or LC-MS detection of dipeptide). Determine K_m for D-Ala at both subsites, K_m for ATP, k_{cat} , and metal (Mg^{2+} vs Mn^{2+}) and K^+ dependence — directly confirming F001/F003.
2. **Product-specificity test.** Assay the purified enzyme with D-alanine vs D-lactate/D-serine as the subsite-2 substrate to confirm it makes the dipeptide and rejects depsipeptide/D-Ser products, testing the prediction from the conserved Tyr253/Ser181/Glu16 triad (F002/F004).
3. **Genetic redundancy test in *P. putida* KT2440.** Construct single (ΔddlA , ΔddlB) and double ($\Delta\text{ddlA} \Delta\text{ddlB}$) deletions; test viability with and without exogenous D-Ala–D-Ala supplementation to confirm the predicted synthetic lethality and redundancy (F005).
4. **D-cycloserine susceptibility.** Determine whether overexpression of PP_4346 raises the DCS MIC of *P. putida* (as seen for *M. smegmatis* Ddl, PMID: 12499203), linking the enzyme to the antibiotic-target pathway.
5. **Structural determination.** Solve the crystal or cryo-EM structure of Q88EV6 (ideally with ATP + a transition-state analog such as a phosphinate), or generate a validated AlphaFold model, to visualize the ATP-grasp fold, the two metal sites, and the ω -loop closure — confirming the mechanistic model.

6. **Inhibitor profiling.** Screen the purified enzyme against known Ddl inhibitor chemotypes (e.g., diazenedicarboxamides, [PMID: 17267218](#)) to assess its tractability as a *Pseudomonas*-relevant antibacterial target.

Conclusion

ddlA (PP_4346, UniProt Q88EV6) of *Pseudomonas putida* KT2440 encodes **cytoplasmic D-alanine–D-alanine ligase A (Ddl, EC 6.3.2.4)**, an ATP-grasp enzyme that condenses two D-alanine molecules into the dipeptide **D-alanyl-D-alanine** ($2 \text{ D-Ala} + \text{ATP} \rightarrow \text{D-Ala-D-Ala} + \text{ADP} + \text{Pi}$), using two $\text{Mg}^{2+}/\text{Mn}^{2+}$ ions and K^+ activation and proceeding through a D-alanyl-phosphate intermediate. This dipeptide is an essential cytosolic peptidoglycan precursor that MurF incorporates into the UDP-MurNAc-pentapeptide. Q88EV6 is a genuine dipeptide (not depsipeptide) ligase — it is 42.8% identical to the structurally defined *E. coli* DdlB and conserves the full catalytic/specificity triad (Glu16/Ser181/Tyr253) and the PROSITE PS00843 signature — and it functions as one of two redundant isozymes (with *ddlB*, Q88N74) in an essential, D-cycloserine-targeted step of bacterial cell-wall biosynthesis.

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